erythrocytes. The erythrocytes will adsorb on to the surface of viral cells multiplying in the culture. This is known as **'haemadsorption'**.

4. Interference

The growth of **non cytopathogenic virus** in cell culture can be tested with the cytopathogenic virus. The growth of the first will inhibit infection by the second virus by **'interference'**.

5. Transformation

Tumour forming viruses **induce cell transformation** and loss of contact inhibition, so that growth appears in a piled up fashion producing **'microtumours'**.

6. Immuno fluorescence

Cells from virus infected cultures are **stained by fluorescent** conjugated antiserum and examined under the UV microscope for the presence of virus antigen.

10.3 Herpes Viruses

The herpes virus family contains more than a hundred species of **enveloped DNA** viruses that affect humans and animals.

Structure

The herpes virus **capsid is icosahedral**, composed of 162 capsomers and enclosing the core containing the linear **double stranded DNA** genome. The nucleocapsid is surrounded by the lipid envelope derived from the host cell. The envelope carries **surface spikes** (Figure 10.4). **Teguments** are present in between the envelope and capsid. The **enveloped virion** measures about 200nm and the **naked virion** about 100 nm in diameter.

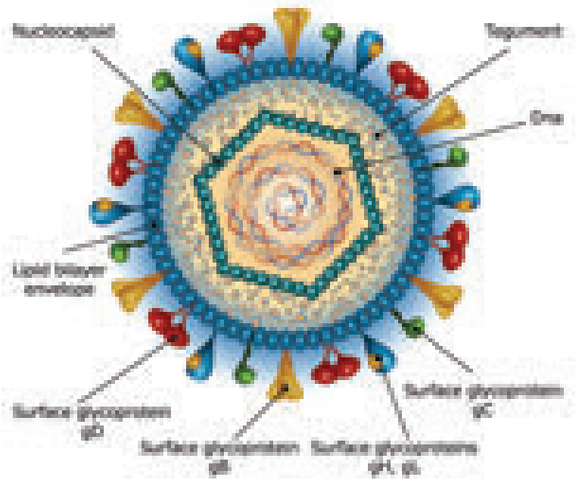


Figure 10.4: Structure of Herpes Simplex Virus

Classification

The family Herpesviridae is divided into **three subfamilies** based on biological, physical and genetic properties (Table 10.1).

Table 10.1: Classification of Human Herpes Viruses

Species	
Name	**Common name**
Human herpes virus type1	Human Simplex virus type1
Human herpes virus type 2	Human Simplex virus type2
Human herpes virus type 3	Varicella Zoster virus
Human herpes virus type 4	Epstein - Barr Virus
Human herpes virus type 5	Cytomegalovirus
Human herpes virus type 6	Human B cell lymphotropic virus*
Human herpes virus type 7	R K Virus *
Human herpes virus type 8	-

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